

ECE 148 Team 6 Final Project

Thermal Patrol

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What we promised

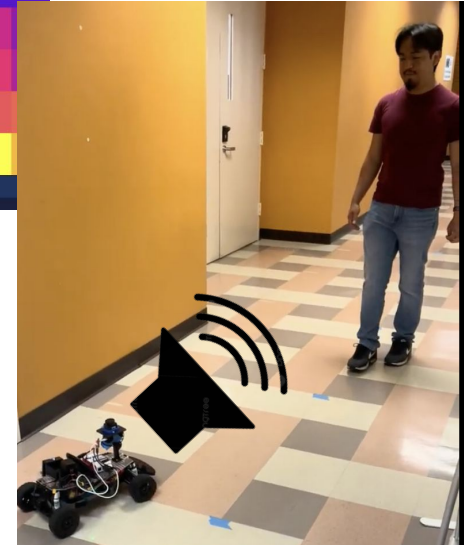
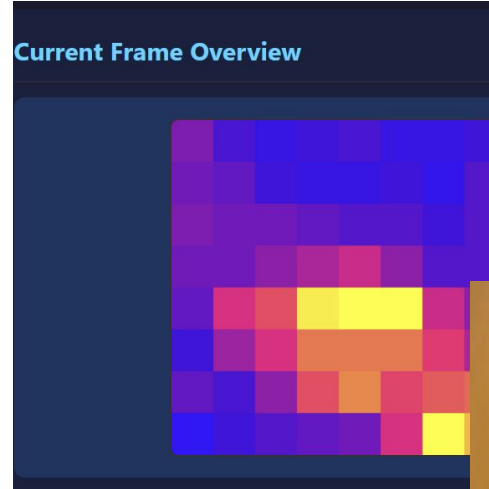
Connect an ESP 32 based thermal camera to the Raspberry Pi via Ros2

Detect a people using the thermal camera, stop, and sound an alarm in response

Navigate autonomously outdoors using GPS and waypoints



Autonomously patrol a hallway using closed loop PID control to center in the corridor

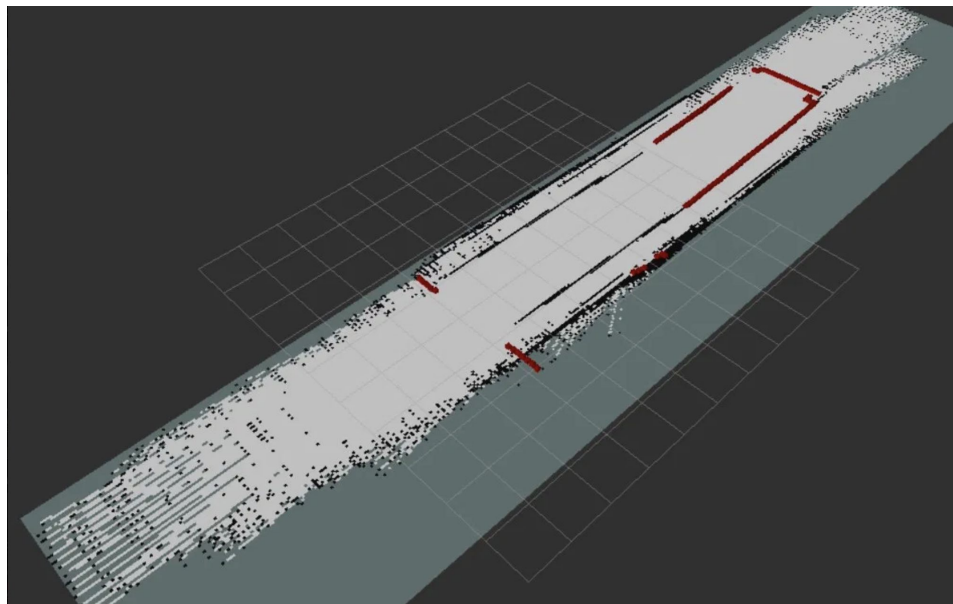


Nice to haves

A web application to notify users of an intruder

Full SLAM based mapping and autonomous navigation using waypoints

Camera integration using a YOLO model to check that the detected object is a person and send a picture to the user



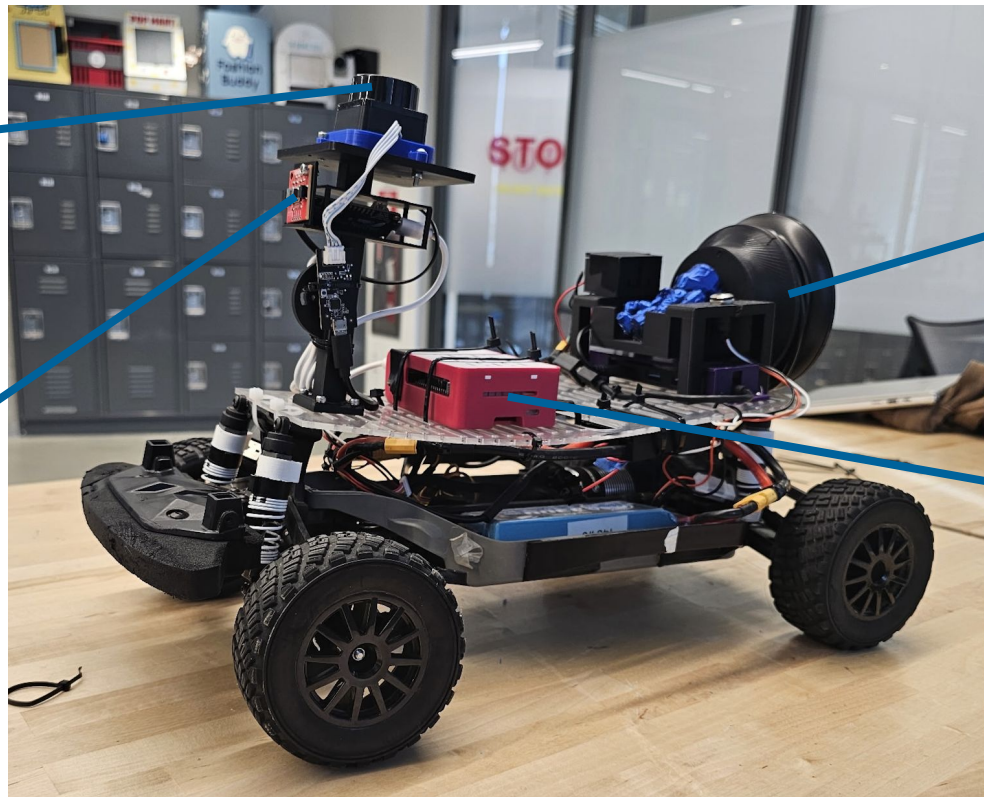
Our Robot

LD19 LiDAR

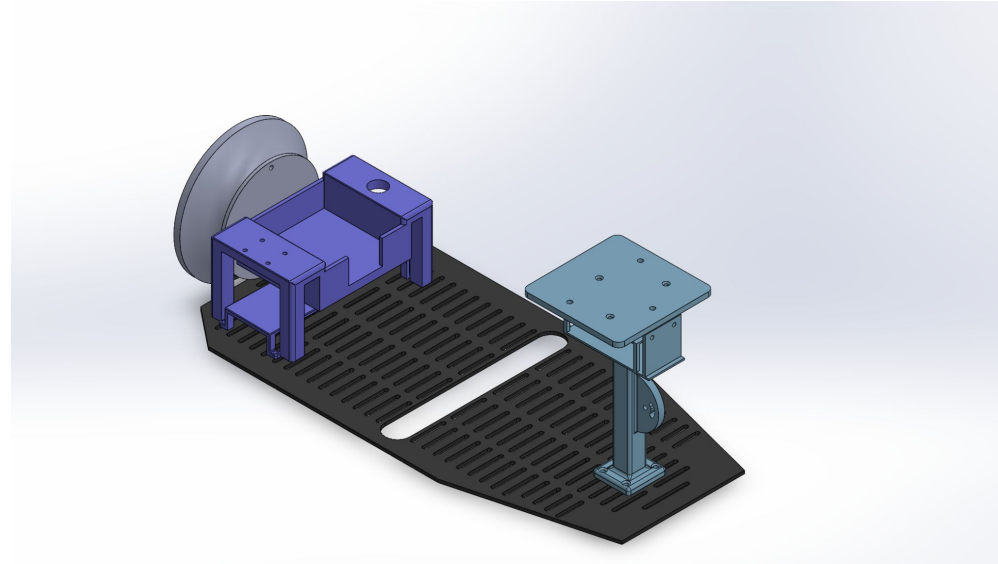
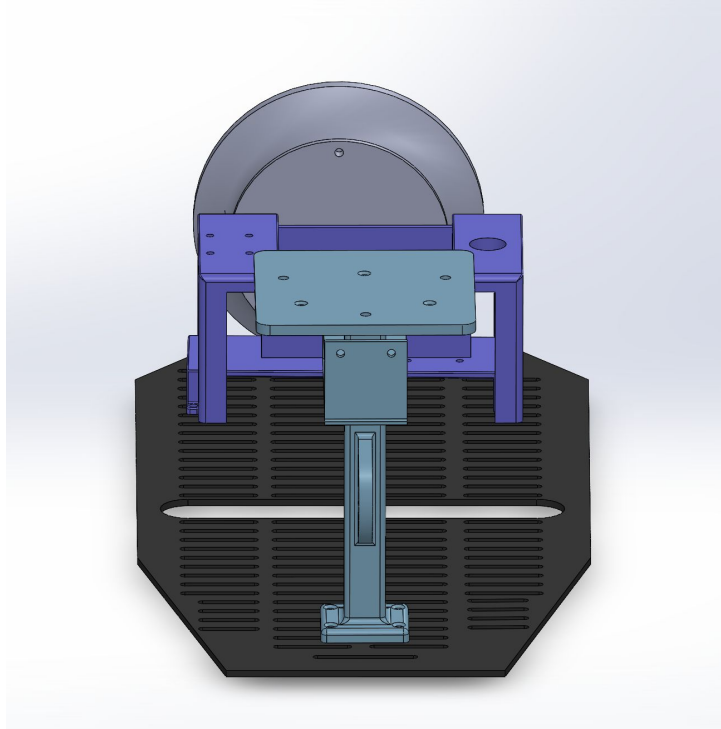
Thermal
Camera

Speaker

Raspberry Pi

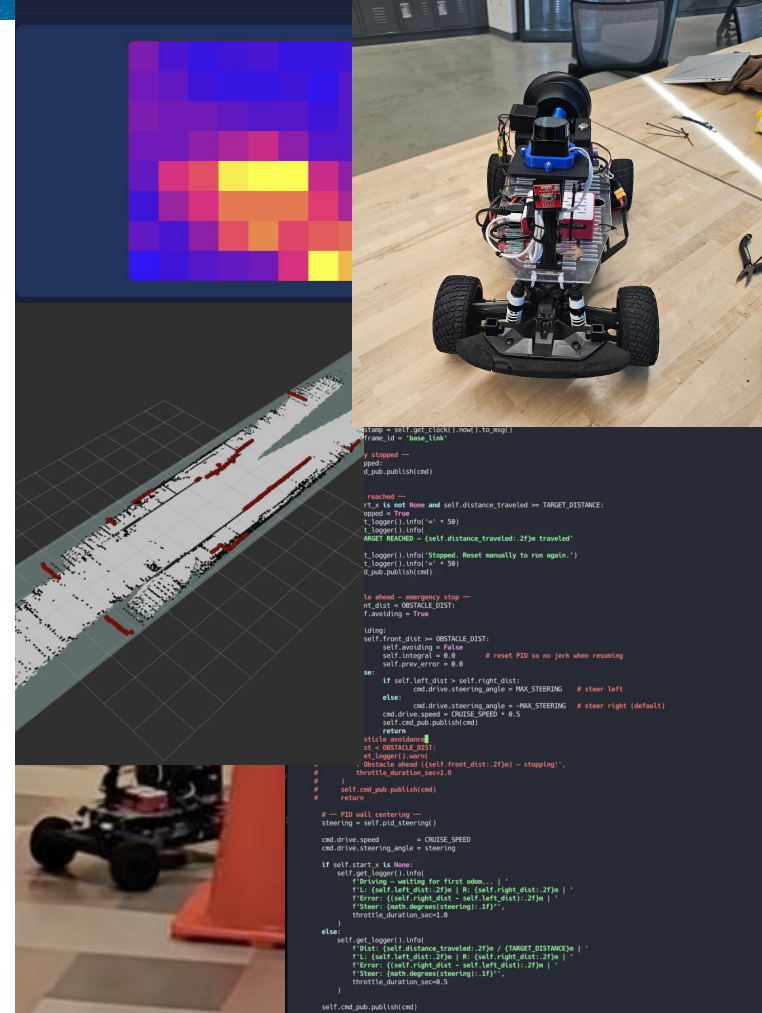


Mechanical Design



What We Have Done

Current Frame Overview



Thermal Detection & MQTT

Thermal Sensing

- An ESP-32 reads in data from the thermal camera and using a custom trained machine learning model to detect heat signatures

MQTT Over Wifi

- The ESP-32 publishes its results over MQTT
- Allows for greater reusability and scaling

Ros 2 Bridge

- A bridge node that subscribes to the ESP-32's MQTT signal and converts that signal into standard ROS2 topics

Autonomous Navigation

PID Control

- LiDAR measures distance from both walls
- Averages the readings to find the center of the hallway
- PID controller continuously steers towards the center of the hallway

Obstacle Avoidance

- Detects obstacles via LiDAR
- Checks for obstacles on either side and turns in the appropriate direction
- Waits till the front is cleared until turning back to the original course

SLAM Mapping

The Challenge

The symmetric hallways lead cause issues with SLAM which relies heavily on pattern matching. With no visual anchors localization drifts and the robot can't properly localize

What We Tried

Increase the confidence threshold and require more readings before accepting a pose estimate, this led to minor improvements but did not fix the underlying issue.

Next Steps

Use RTAB mapping and integrate an OAK-D camera to provide visual anchors to allow for more accurate mapping

Videos





What didn't work

GPS

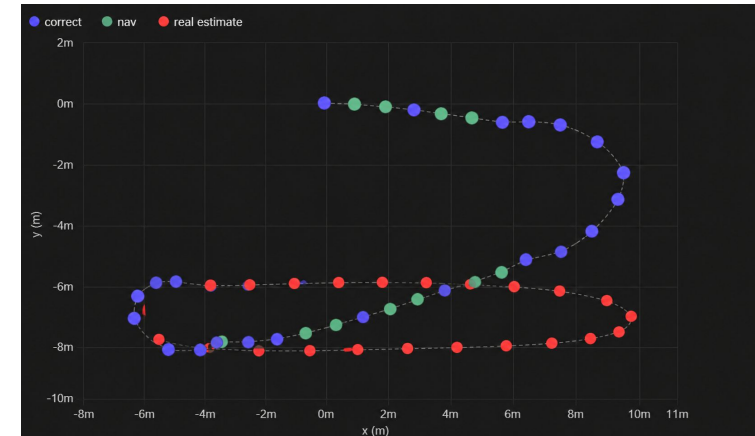
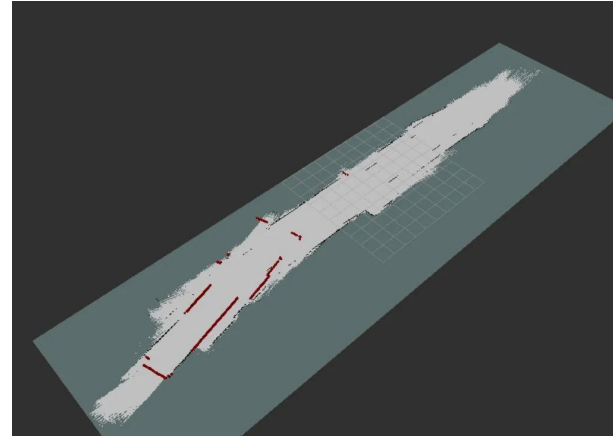
- Unreliable signal
- Inaccurate readings

Odometry

Significant drift in our robot's odometry readings made it unreliable and impacting our SLAM's mapping

SLAM

- Generated noisy and inconsistent maps
- Unable to consistently map a symmetrical hallway



Solutions to our problems

GPS

- Determined that our inaccuracies came down to hardware
- Pivoted to an indoor LiDAR based solution

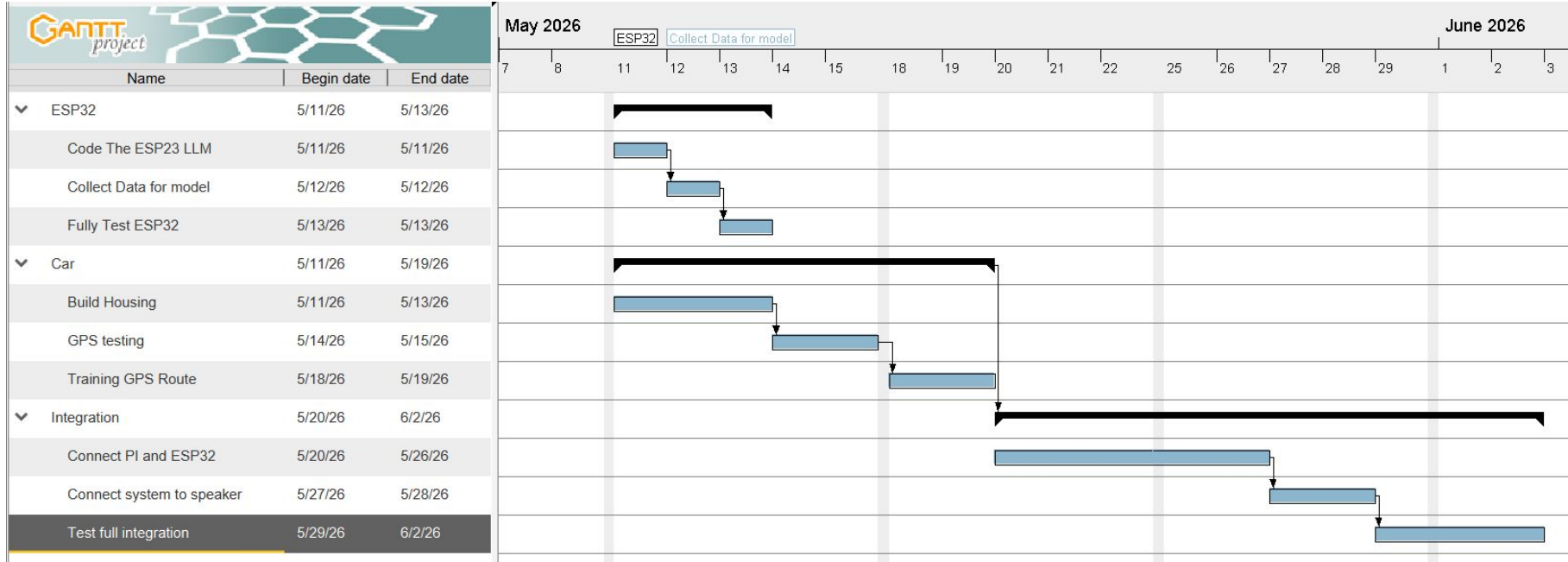
Odometry

- Relied more heavily on LiDAR based solutions
- Closed loop control based on the LiDAR's measurements

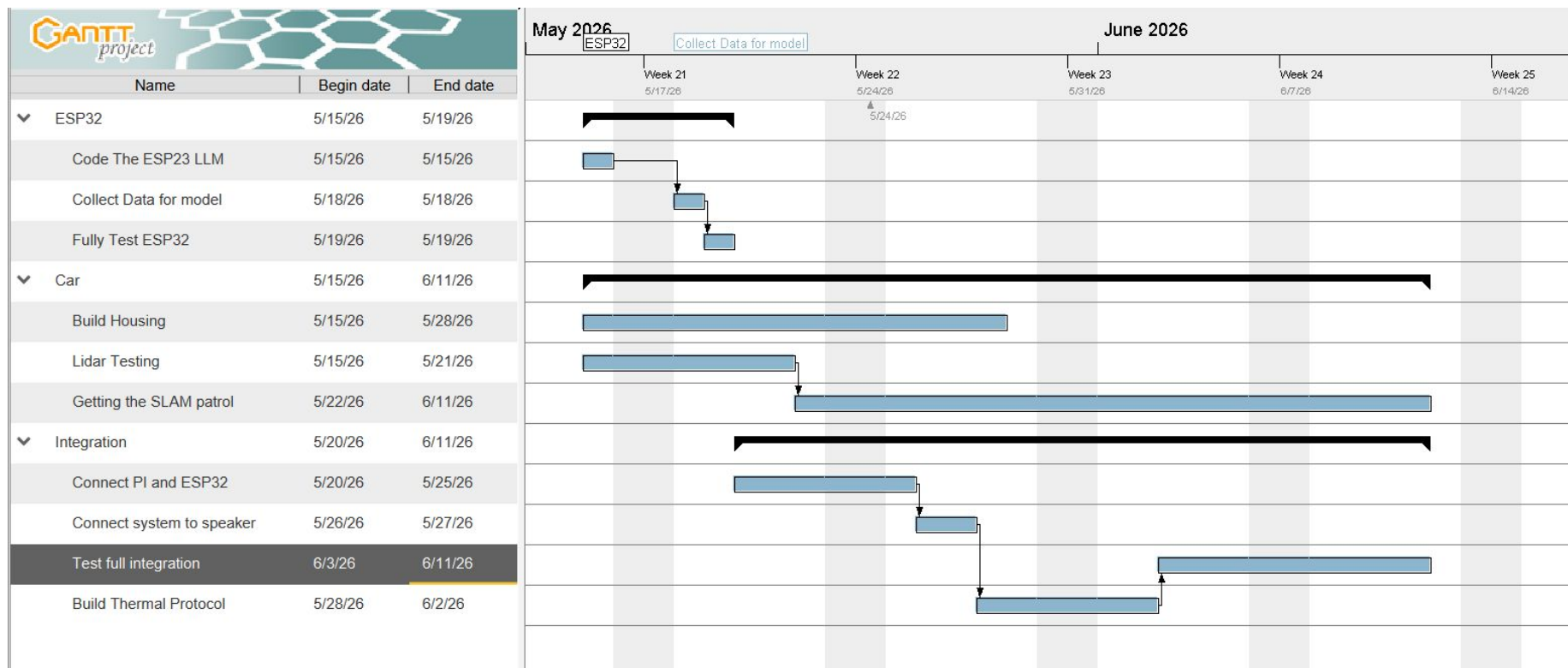
SLAM

- Tried filtering out bad poses
- If time permitted would have implemented a RTAB based localization system, adding an OAK-D camera to our measurements

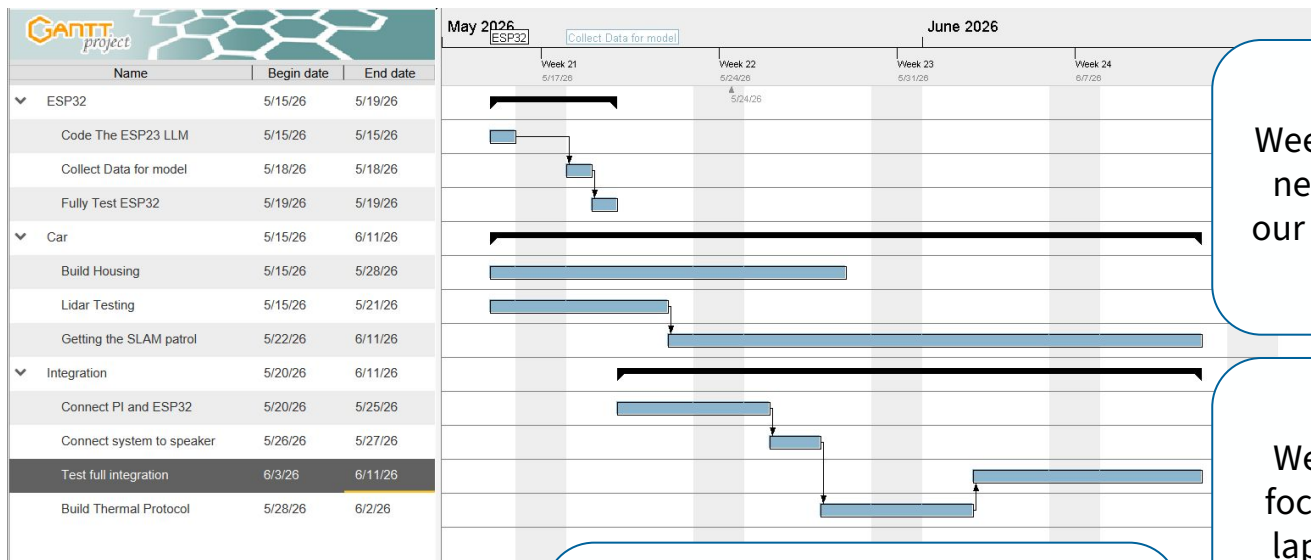
Original Gantt chart



Updated Gantt Chart



Updated Gantt Chart - Lessons Learned



Changes

Week 8 or so we decided that it was necessary to expand the scope of our project from using GPS to LiDAR for indoor navigation

Challenges

We spent much of our first weeks focusing on finishing up the ROS 2 laps and debugging software and hardware issues.

Lessons Learned

- Start early and test frequently
- Expect things to break
- Leave time for debugging
- Research before jumping to a solution (e.g. SLAM)

If we had another week

Integrate RTAB Mapping

- Combine our existing LiDAR with an OAK-D camera to improve mapping in symmetrical hallways using visual landmarks
- Corrects for drift and current SLAM failure modes

Better Thermal Camera

- Our current thermal camera has very low resolution allowing for misidentification of objects or animals as people

Create a webapp

- Send notifications to users to create a more active autonomous 'security guard'
- Allow them to use the web app to control the robot, draw maps, start patrol ect

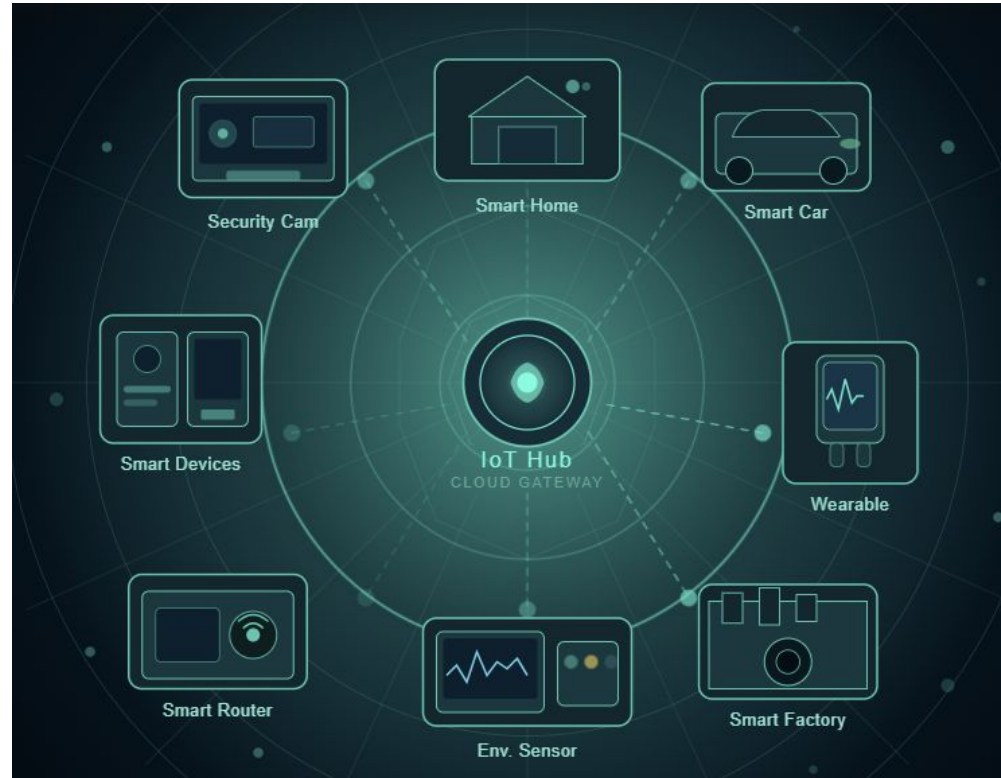
Integrate a YOLO model

- Add a camera and light to run a YOLO model to detect to verify that an object is infact a person and send a notification to the user

For those who come after

Ideas for expanding

- MQTT doesn't require physical connections
- Communication between different robots and sensors that are not plugged in together is possible



Thank you!